

Date	31 January 2025	Team ID	6ab61594fa5e5219dc387253
Project Name	Import Data using Transform Maps	Maximum Marks	4 Marks

PROJECT DESIGN PHASE-II | SOLUTION REQUIREMENTS

Solution Requirements (Functional & Non-functional)

Functional Requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Source & Staging Load	- Legacy CSV file upload interface. - Scheduled import data source configuration. - Staging table data ingestion ('u_training_staging').
FR-2	Transform Map & Coalesce Matching	- Unique key coalesce matching (Employee ID / Email). - Field mapping rules for training score and completion date. - Handling duplicate records and staging error logs.
FR-3	Reference Resolution via Dot-Walking	- Dynamic department and manager reference lookups. - Scripted field mapping for dot-walking relationships. - Validation and error handling for unresolved parent references.
FR-4	Role-Based Access Control & ACLs	- Custom role creation ('x_training_admin', 'x_training_viewer'). - Table-level Read/Write access control lists. - Field-level security for confidential training score results.

Non-functional Requirements:

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Intuitive form layouts, clear error messaging during transform map execution, and guided admin interfaces for data management.
NFR-2	Security	Strict enforcement of role-based access control (RBAC), SHA-256 encrypted data transmission, and audit logging for record modifications.
NFR-3	Reliability	Transaction rollback on transform script failures and robust data validation checks prior to target table ingestion.
NFR-4	Performance	Optimized batch processing for large legacy CSV files (up to 10,000 records) completing transform execution within 3 minutes.
NFR-5	Availability	99.9% instance uptime backed by ServiceNow multi-tenant cloud infrastructure and high-availability server clustering.
NFR-6	Scalability	Ability to scale import worker processing horizontally via ServiceNow background job scheduler and asynchronous queues.